P1

$$\dot{x}_1 = (2 - x_1 - x_2)\, x_1$$
$$\dot{x}_2 = (x_1 - x_2)\, x_2$$

To find equilibrium pts x_e and local stability.

Let $x_e = (x_{1e}, x_{2e})$ satisfies $f_1(x_{1e}, x_{2e}) = 0$ & $f_2(x_{1e}, x_{2e}) = 0$

$\therefore (2 - x_1 - x_2)\, x_1 = 0$

or $x_1 = 0$ or $2 - x_1 - x_2 = 0$

$\therefore x_2 = 2 - x_1$ or $x_1 + x_2 = 2$

& $(x_1 - x_2)\, x_2 = 0$ $\therefore x_2 = 0$ or $x_1 = x_2$

when $x_1 = 0$,

if $x_2 = 0$, $\rightarrow x_{1e_0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

if $x_1 = x_2$, $x_2 = 0$

when $x_1 + x_2 = 2$

if $x_2 = 0 \rightarrow x_1 = 2$, $x_{2e} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

if $x_1 = x_2 \neq$, $x_1 = 1$, $\&\ x_2 = 1$

$\therefore x_{3e} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore$ the three equilibrium points are

$$x_{1e} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_{2e} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad x_{3e} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

To determine stability

$$D f(x_1, x_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial}{\partial x_1}(2x_1 - x_1^2 - x_2 x_1) & \frac{\partial}{\partial x_2}(2x_1 - x^2 - x_2 x_1) \\ \frac{\partial}{\partial x_1}(x_1 x_2 - x_2^2) & \frac{\partial}{\partial x_2}(x_1 x_2 - x_2^2) \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 2x_1 - x_2 & -x_1 \\ x_2 & x_1 - 2x_2 \end{bmatrix}$$

at $(x_{1e}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad Df = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ UNSTABLE

2×2 upper triangular it has eigen value 2 and 0

since one $\lambda_1 = 2 > 0$ i.e. positive real part so system is unstable at $x_{1e} = (0,0)$ locally and grows exponentially as e^{2t}.

at $(x_{2e}) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $\quad Df(2,0) = \begin{bmatrix} -2 & -2 \\ 0 & 2 \end{bmatrix}$ UNSTABLE

eigen values are $\lambda_1 = -2$ and $\lambda_2 = 2$

here for $\lambda_1 = -2 < 0$ it is locally stable] none lies on $Re(\lambda) \neq 0$

but $\lambda_2 = 2 > 0$ it's locally unstable. so HG theorem applies.

So its "unstable" (saddlept) case as per notes-lectures.

at $(x_{3e}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\quad Df(1,1) = \begin{bmatrix} 2-2-1 & -1 \\ 1 & 1-2 \end{bmatrix}$

$= \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$ STABLE

$\det(A - \lambda I) = 0$

$\begin{vmatrix} -1-\lambda & 1 \\ -1 & -1-\lambda \end{vmatrix} = 0$ $\quad (-(1-\lambda)^2 + 1 = 0$ or $\lambda^2 + 2\lambda + 2 = 0$

$\therefore (\lambda+1)^2 = -1$

$\lambda + 1 = \pm i$

$\lambda_1 = -1 + i$ $\quad$ & $\lambda_2 = -1 - i$

here both λ_1 & λ_2 has $-1 < 0$ negative real part so it is stable since $Re(\lambda_i) < 0$

$\therefore x_{3e} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is locally asymptotically stable.

(P2)

$$\dot{x}_1(t) = x_2(t) - x_1(t)$$

$$\dot{x}_2(t) = \frac{x_1(t) + x_3(t)}{2} - x_2(t)$$

$$\dot{x}_3(t) = x_2(t) - x_3(t) + u(t)$$

dynamics $\dot{x}(t) = Ax(t) + Bu(t)$

where $A = \begin{bmatrix} -1 & 1 & 0 \\ 1/2 & -1 & 1/2 \\ 0 & 1 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

To calculate eigen value

$$\det(-A + \lambda I) = 0$$

$$\begin{vmatrix} \lambda+1 & -1 & 0 \\ -1/2 & \lambda+1 & -1/2 \\ 0 & -1 & \lambda+1 \end{vmatrix} = 0 \quad \text{or} \quad (\lambda+1)\left[(\lambda+1)^2 - \tfrac{1}{2}\right] - 1\left[\tfrac{1}{2}(\lambda+1)\right] = 0$$

or $(\lambda+1)\left[\lambda^2 + 2\lambda + 1 - \frac{1}{2}\right] - \frac{1}{2}(\lambda+1) = 0$

or $(\lambda+1)\left[\lambda^2 + 2\lambda + 1 - 1\right] = 0$

or $(\lambda+1)\lambda(\lambda+2) = 0$ $\therefore$ $\lambda_1 = -1$, $\lambda_2 = 0$, $\lambda_3 = -2$

right eigen vector.

$\lambda_1 = -1$, $\begin{bmatrix} -1+1 & -1 & 0 \\ -1/2 & -1+1 & -1/2 \\ 0 & -1 & -1+1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$ $\quad \begin{bmatrix} 0 & -1 & 0 \\ -1/2 & 0 & -1/2 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$

$v_2 = 0$, $-1/2\, v_1 + \frac{1}{2} v_3 = 0$ $\quad v_2 = 0$

$v_1 = -v_3$ $\quad \therefore \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$\lambda_2 = 0$, $\begin{bmatrix} 1 & -1 & 0 \\ -1/2 & 1 & -1/2 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$v_1 + v_2 = 0$ $\quad -1/2\, v_1 + v_2 - 1/2\, v_3 = 0$ $\quad -v_2 + v_3 = 0$

$v_1 = -v_2$ $\quad v_1 = v_2 = v_3$ $\quad \therefore \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\quad v_2 = -v_3$

$\lambda_3 = -2$

$$\begin{bmatrix} -1 & -1 & 0 \\ -\frac{1}{2} & -1 & -\frac{1}{2} \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$v_1 = v_2$ | $-\frac{1}{2}v_1 - v_2 - \frac{1}{2}v_3 = 0$ $\quad -v_2 - v_3 = 0$

$v_2 = -v_3$

$v_1 = 1, \quad v_2 = -1, \quad v_3 = +1$

$$\therefore \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Left ones $\quad \omega_i^T A = \lambda_i \omega_i^T$ & normalize

$\lambda_1 = -1$, $\quad \begin{bmatrix} \omega_{11} & \omega_{12} & \omega_{13} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$\omega_{12} = 0 \quad \therefore \omega_{12} = 0$

$\omega_{11} + \omega_{13} = 0 \quad \omega_{11} = -\omega_{13}$

$\frac{1}{2}\omega_{12} - \omega_{13} = 0 \quad \therefore$

$\omega_1^T = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$

$\omega_1^T v_1 = 1$

$\alpha \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 1$ or $\alpha = \frac{1}{2}$

$\therefore \omega_1^T = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix}$

$\lambda_2 = 0 \quad \begin{bmatrix} \omega_{21} & \omega_{22} & \omega_{23} \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ \frac{1}{2} & -1 & \frac{1}{2} \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$-\omega_{21} + \frac{1}{2}\omega_{22} = 0, \quad \omega_{22} = 2\omega_{21}$

$\omega_{21} - \frac{1}{2}\omega_{22} + \omega_{23} = 0 \quad \omega_{22} = 2\omega_{23}$

$\frac{1}{2}\omega_{22} - \omega_{23} = 0$

$\therefore \omega_2^T = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$

$\omega_2^T v_2 = 1$

$\beta \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 1 \quad 4\beta = 1 \quad \beta = \frac{1}{4}$

$\checkmark \omega_2^T = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$

$\lambda_3 = -2 \quad \begin{bmatrix} \omega_{31} & \omega_{32} & \omega_{33} \end{bmatrix} \begin{bmatrix} -1 & -1 & 0 \\ -\frac{1}{2} & -1 & -\frac{1}{2} \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$-\omega_{31} - \frac{1}{2}\omega_{32} = 0 \quad \omega_{31} = -\frac{1}{2}\omega_{32} \quad \omega_{32} = -2\omega_{31}$

$-\frac{1}{2}\omega_{32} - \omega_{33} = 0 \quad -2\omega_{33} = \omega_{32} \quad \omega_3^T = \begin{bmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \end{bmatrix}$

$\omega_3^T = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$ ✓

General soln

$$PAP^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & -1/2 \\ 1/4 & 1/2 & -1/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}$$

$$e^{At} = P e^{\Lambda t} P^{-1}$$

$$= \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}}_{P} \underbrace{\begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix}}_{e^{\Lambda t}} \underbrace{\begin{bmatrix} 1/2 & 0 & -1/2 \\ 1/4 & 1/2 & 1/4 \\ -1/4 & -1/2 & 1/4 \end{bmatrix}}_{P^{-1}}$$

- $x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau)\, d\tau$

$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ & $u(t) \equiv 1$

$$\therefore x(t) = e^{At}x(0) + P \int_0^t \begin{bmatrix} e^{-(t-\tau)} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} -1/2 \\ +1/4 \\ 1/4 \end{bmatrix} d\tau \quad \text{—(1)}$$

$$\int_0^t e^{-(t-\tau)}\, d\tau = 1 - e^{-t} \qquad \int_0^t d\tau = t \qquad \int_0^t e^{-2(t-\tau)}\, d\tau$$

$$= \tfrac{1}{2} e^{-2t+2\tau} \Big]_0^t$$

$$= 1/2\, e^{-2t+2t} - \tfrac{1}{2} e^{-2t}$$

$$= \frac{1 - e^{-2t}}{2}$$

$$\int_0^t e^{A(t-\tau)} B\, d\tau$$

$$= P \begin{bmatrix} -\frac{1}{2}(1-e^{-t}) \\ 1/4 \cdot t \\ 1/4 \left(\frac{1-e^{-2t}}{2}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1/2 (1-e^{-t}) \\ t/4 \\ \frac{1}{8}(1-e^{-2t}) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}(1-e^{-t}) + \frac{t}{4} + \frac{1}{8}(1-e^{-2t}) \\ t/4 - \frac{1}{8}(1-e^{-2t}) \\ 1/2(1-e^{-t}) + t/4 + \frac{1}{8}(1-e^{-2t}) \end{bmatrix}$$

$$= \begin{bmatrix} t/4 - 3/8 + \frac{1}{2}e^{-t} - \frac{1}{8}e^{-2t} \\ t/4 - 1/8 + \frac{1}{8}e^{-2t} \\ t/4 + 5/8 - 1/2\, e^{-t} - \frac{1}{8}e^{-2t} \end{bmatrix}$$

also now $P^{-1}x(0) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \frac{x_1(0) - x_3(0)}{2} \\ \frac{x_1(0) + 2x_2(0) + x_3(0)}{4} \\ \frac{x_1(0) - 2x_2(0) + x_3(0)}{4} \end{bmatrix}$

$$\therefore e^{At}x(0) = P\begin{bmatrix} e^{-t}q_1 \\ q_2 \\ e^{-2t}q_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t}q_1 + q_2 + e^{-2t}q_3 \\ q_2 - e^{-2t}q_3 \\ -e^{-t}q_1 + q_2 + e^{-2t}q_3 \end{bmatrix}$$

$$q = P^{-1}x(0) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

$$e^{At}q = \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

$$\therefore x(t) = \begin{bmatrix} e^{-t}q_1 + q_2 + e^{-2t}q_3 \\ q_2 - e^{-2t}q_3 \\ -e^{-t}q_1 + q_2 + e^{-2t}q_3 \end{bmatrix} + \begin{bmatrix} t/4 - 3/8 + \frac{1}{2}e^{-t} - \frac{1}{8}e^{-2t} \\ t/4 - 1/8 + 1/8\,e^{-2t} \\ t/4 + \frac{5}{8} - \frac{1}{2}e^{-t} - \frac{1}{8}e^{-2t} \end{bmatrix}$$

$e^{At}x(0)$ ↑ + $\int_0^t e^{A(t-\tau)}B\,u(\tau)\,d\tau$ ↑

This above can form an analytical expression

$$= \begin{bmatrix} t/4 + q_2 - 3/8 + e^{-t}\left(q_1 + \frac{1}{2}\right) + e^{-2t}\left(q_3 - 1/8\right) \\ q_2 + t/4 - \frac{1}{8} + e^{-2t}\left(\frac{1}{8} - q_3\right) \\ q_2 + t/4 + 5/8 - e^{-t}\left(q_1 + \frac{1}{2}\right) + e^{-2t}\left(q_3 - 1/8\right) \end{bmatrix}$$

Ans

as $t \to \infty$, $e^{-t} \to 0$ and $e^{-2t} \to 0$

$\therefore x_1(t) \approx q_2 + t/4 - 3/8$, $x_2(t) \approx q_2 + t/4 - 1/8$, $x_3(t) \approx q_2 + t/4 + 5/8$

$(x_1 - x_2) \to (q_2 + t/4 - 3/8) - (q_2 + t/4 - 1/8) = -3/8 + \frac{1}{8} = -\frac{1}{4}$

$\boxed{|x_1 - x_2| = 1/4}$

$(x_2 - x_3) \to (q_2 + t/4 - \frac{1}{8}) - (q_2 + t/4 + 5/8) = -3/4$ $\boxed{|x_2 - x_3| = \frac{3}{4}}$

$\therefore$ the distances $|x_1 - x_2|$ & $|x_2 - x_3|$ are $1/4$ & $3/4$

Ans

2(b) $u(t) = \sin t$ for $t \geq 0$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & 1 & 0 \\ 1/2 & -1 & 1/2 \\ 0 & 1 & -1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1/2 & 0 & -1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix} \quad P^{-1}B = \begin{bmatrix} -1/2 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$x(t) = e^{At}x(0) + P \int_0^t \begin{bmatrix} e^{-(t-\tau)} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-2(t-\tau)} \end{bmatrix} P^{-1}B \sin\tau \, d\tau \quad \text{---(1)}$$

The integrals I become

$$P \begin{bmatrix} -1/2 \int_0^t e^{-(t-\tau)} \sin\tau \, d\tau \\ 1/4 \int_0^t \sin\tau \, d\tau \\ 1/4 \int_0^t e^{-2(t-\tau)} \sin\tau \, d\tau \end{bmatrix} = P \begin{bmatrix} -1/2 \, I_1 \\ 1/4 \, I_2 \\ 1/4 \, I_3 \end{bmatrix}$$

$$I_1 = \int_0^t e^{-(t-\tau)} \sin\tau \, d\tau$$

$$= \frac{\sin t - \cos t - e^{-t}}{2}$$

$$I_2 = \int_0^t \sin\tau \, d\tau = 1 - \cos t$$

$I_3 =$

$$= \int_0^t e^{-2(t-\tau)} \sin\tau \, d\tau$$

$$= \frac{1}{2}\sin t - \frac{1}{2}\int_0^t e^{-2(t-\tau)} \cos\tau \, d\tau$$

$$= \frac{1}{2}\sin t - \frac{1}{2}\left[\left(\cos\tau \cdot \frac{1}{2} e^{-2(t-\tau)}\right]_0^t - \frac{1}{2}\int_0^t e^{-2(t-\tau)} (-\sin\tau) \, d\tau\right]$$

$$= \frac{1}{2}\sin t - \frac{1}{2}\left[\left(\frac{1}{2}\cos t - \frac{1}{2}e^{-2t}\right) + \frac{1}{2} I_3\right]$$

$$\frac{5}{4} I_3 = \frac{1}{2}\sin t - \frac{1}{4}\cos t + \frac{1}{4} e^{-2t}$$

$$\therefore I_3 = \frac{2\sin t - \cos t + e^{-2t}}{5}$$

let $I_1 = \int_0^t e^{-(t-\tau)} \sin\tau \, d\tau$

$\int u \, dv = uv - \int v \, du$

$u = \sin\tau \quad du = \cos\tau \, d\tau$

$$\therefore \left[\sin\tau \, e^{-(t-\tau)}\right]_0^t - \int_0^t e^{-(t-\tau)} \cos\tau \, d\tau$$

$$= \left[\sin t - \int_0^t e^{-(t-\tau)} \cos\tau \, d\tau\right]$$

$$= \sin t - \left[\left[\cos\tau \, e^{-(t-\tau)}\right]_0^t - \int_0^t e^{-(t-\tau)} (-\sin\tau) \, d\tau\right]$$

$$= \sin t - \left[(\cos t - e^{-t}) + \int_0^t e^{-(t-\tau)} \sin\tau \, d\tau\right]$$

$$I_1 = \sin t - \left[(\cos t - e^{-t}) + I_1\right]$$

$$2I_1 = \sin t - \cos t - e^{-t}$$

$$I_1 = \frac{\sin t - \cos t - e^{-t}}{2}$$

$\therefore$ ~~Proof~~ $e^{A(t-\tau)} B \sin\tau$ ~~$\sin t - \frac{1}{20}\cos t$~~

$r_1 = -\frac{1}{2} I_1$, $r_2 = \frac{1}{4} I_2$ $r_3 = \frac{1}{4} I_3$ $\therefore P \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} r_1 + r_2 + r_3 \\ r_2 - r_3 \\ -r_1 + r_2 + r_3 \end{bmatrix}$

$$q_1 + q_2 + q_3 = -\frac{1}{2} I + \frac{1}{4} I_2 + \frac{1}{4} I_3 = -\left(\frac{\sin t - \cos t + e^{-t}}{4}\right) + \frac{1 - \cos t}{4} + \frac{2\sin t - \cos t + e^{-2t}}{20}$$

$$= \frac{1}{4} - \frac{1}{20}\cos t - \frac{3}{20}\sin t - \frac{1}{4} e^{-t} + \frac{1}{20} e^{-2t}$$

$$r_2 - r_3 = \frac{1 - \cos t}{4} - \frac{2\sin t - \cos t + e^{-2t}}{20}$$

$$= \frac{1}{4} - \frac{1}{5}\cos t - \frac{1}{10}\sin t - \frac{1}{20} e^{-2t}$$

$$-r_1 + r_2 + r_3 = \frac{+\sin t - \cos t + e^{-t}}{4} + \frac{1 - \cos t}{4} + \frac{2\sin t - \cos t + e^{-2t}}{20}$$

$$= \frac{1}{4} - \frac{11}{20}\cos t + \frac{7}{20}\sin t + \frac{1}{4} e^{-t} + \frac{1}{20} e^{-2t}$$

eq $\Rightarrow$ ①

$$x(t) = e^{At} \cdot x(0) + \int_0^t e^{A(t-\tau)} B \sin\tau \, d\tau$$

$$= \begin{bmatrix} e^{-t} q_1 + q_2 + e^{-2t} q_3 \\ q_2 - e^{-2t} q_3 \\ e^{-t} q_1 + q_2 + e^{-2t} q_3 \end{bmatrix} + \begin{bmatrix} 1/4 - \frac{1}{20}\cos t - \frac{3}{20}\sin t + \frac{e^{-t}}{4} + \frac{1}{20} e^{-2t} \\ 1/4 - \frac{1}{5}\cos t - \frac{1}{10}\sin t - \frac{1}{20} e^{-2t} \\ \frac{1}{4} - \frac{11}{20}\cos t + \frac{7}{20}\sin t - \frac{1}{4} e^{-t} + \frac{1}{20} e^{-2t} \end{bmatrix}$$

from part (a)

as $t \to \infty$, exponential term $\to 0$

$\therefore$

$$= \begin{bmatrix} q_2 + \frac{1}{4} - \frac{1}{20}\cos t - \frac{3}{20}\sin t \\ q_2 + \frac{1}{4} - \frac{1}{5}\cos t - \frac{1}{10}\sin t \\ q_2 + \frac{1}{4} - \frac{11}{20}\cos t + \frac{7}{20}\sin t \end{bmatrix}$$

~~Answer~~ Answer ?

Amplitudes ~~Ans~~

$$\text{Amp1} = \sqrt{\left(-\frac{1}{20}\right)^2 + \left(-\frac{3}{20}\right)^2} = \frac{\sqrt{10}}{20}$$

~~Answer~~

$$\text{Amp2} = \sqrt{\left(-\frac{1}{5}\right)^2 + \left(-\frac{1}{10}\right)^2} = \frac{\sqrt{20}}{20}$$

$$\text{amp3} = \sqrt{\left(\frac{11}{20}\right)^2 + \left(\frac{7}{20}\right)^2} = \frac{\sqrt{170}}{20}$$

(8a)

let's assume $a_2 = 0$ as Matlab Code uses $x(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\therefore \frac{x_1(0) + 2x_2(0) + x_3(0)}{4} = 0 .$$

$$\therefore x_1(t) \approx \frac{1}{4} - \frac{1}{20}\cos t - \frac{3}{20}\sin t$$

$$x_2(t) \approx \frac{1}{4} - \frac{1}{5}\cos t - \frac{1}{10}\sin t$$

$$x_3(t) \approx \frac{1}{4} - \frac{11}{20}\cos t + \frac{7}{20}\sin t$$

Answer

Matlab shows steady state motion

$$x_1(t) \approx \frac{1}{4} + 0.1581 \sin(t - 161.57^\circ)$$

$$x_2(t) \approx \frac{1}{4} + 0.2236 \sin(t - 116.57^\circ)$$

$$x_3(t) \approx \frac{1}{4} + 0.6519 \sin(t - 57.53^\circ)$$

For $x(0)$ choice of 0 here, their amps are $\frac{\sqrt{10}}{20}$, $\frac{\sqrt{20}}{20}$ & $\frac{\sqrt{170}}{20}$

ie 0.16, 0.22 & 0.65 & phase shifts are -161.57°, -116.57° & -57.53°. & mean pos ~ 0.25 (Answer)

So car 3 has max amp $\overset{0.65}{\wedge}$ because it gets the external force $u(t)$ and Car 3 responds first & with small phase delay (-57°) & Car 2 & Car 3 follows & with -117° & 162° respectively.

Answer.

```
clear;
clc;
close all;

% System: x_dot = A*x + B*u, with u(t) = sin(t)
A = [-1    1     0;
    1/2 -1    1/2;
    0    1    -1];

B = [0; 0; 1];
Cmat = eye(3);
D = zeros(3,1);

sys = ss(A, B, Cmat, D);

% Initial positions
x0 = [0; 0; 0];

% Time and input
t = (0:0.01:40)';
u = sin(t);

% Simulate
[y, tout] = lsim(sys, u, t, x0);

% Steady-state form
q2 = (x0(1) + 2*x0(2) + x0(3))/4;
meanPosition = q2 + 1/4;

Ccoef = [-1/20; -1/5; -11/20];
Scoef = [-3/20; -1/10; 7/20];

% Amplitude and phase
amplitude = hypot(Ccoef, Scoef);
phase_deg = atan2d(Ccoef, Scoef);

% Print results
car = ["Car 1"; "Car 2"; "Car 3"];
Results = table(car, repmat(meanPosition,3,1), amplitude, ...
    phase_deg, ...
    'VariableNames', {'Car','MeanPosition','Amplitude', ...
    'Phase_deg',});
disp(Results);
```

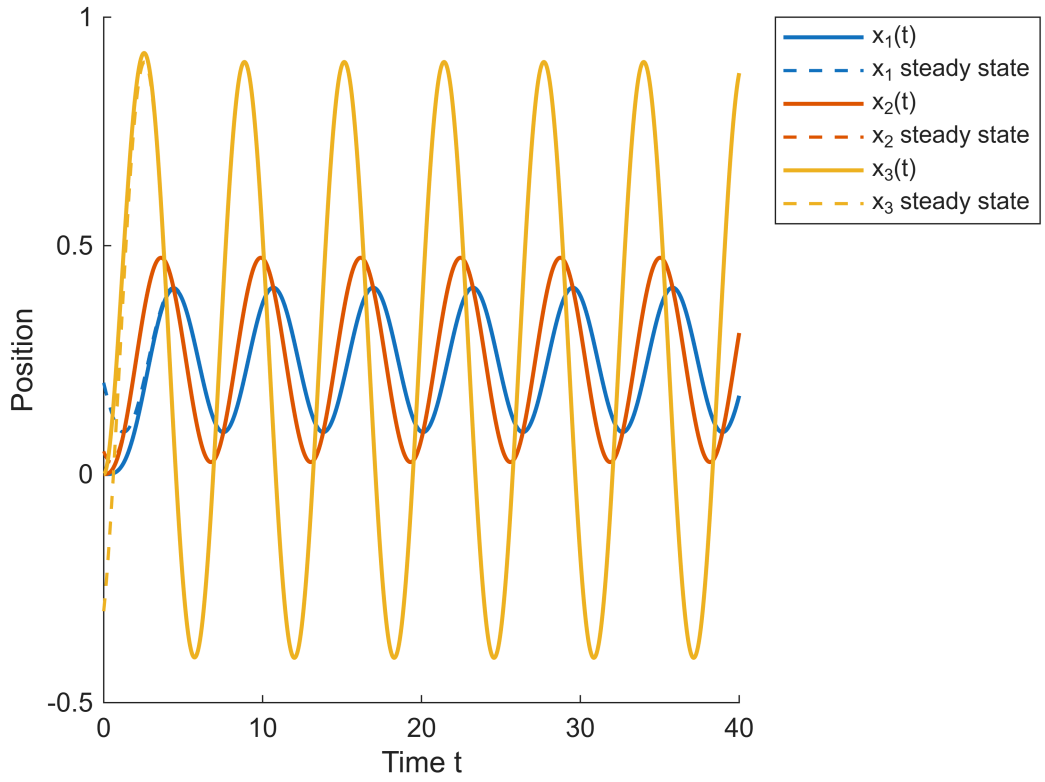
Car	MeanPosition	Amplitude	Phase_deg
"Car 1"	0.25	0.15811	-161.57
"Car 2"	0.25	0.22361	-116.57
"Car 3"	0.25	0.65192	-57.529

```
% Calculate steady-state curves
yss = meanPosition + cos(tout)*Ccoef.' + sin(tout)*Scoef.';

% Plot simulated positions
colors = lines(3);
figure('Color','w');
hold on;

legendText = cell(1,6);
for i = 1:3
    plot(tout, y(:,i), 'Color', colors(i,:), 'LineWidth', 1.5);
    plot(tout, yss(:,i), '--', 'Color', colors(i,:), 'LineWidth', 1.2);

    legendText{2*i-1} = sprintf('x_%d(t)', i);
    legendText{2*i}   = sprintf('x_%d steady state', i);
end
grid off;
xlabel('Time t');
ylabel('Position');
legend(legendText, 'Location', 'bestoutside');
```



(P3) $\dot{x}(t) = A(t)\,x(t) + Bu(t) = \begin{bmatrix} -1 & 2t \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$ (9)

$x(0) \in \mathbb{R}^2$

(a) $A(s) = -I + sN \qquad N = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \; N^2 = 0$

$A(s)\,A(t) = (-I + 2sN)(-I + 2tN) = I - 2(s+t)N = A(t)\,A(s)$ ↑

$\therefore$ they commute

$= \begin{bmatrix} 1 - 2(s+t) & \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-2(t+s) & \\ 0 & 1 \end{bmatrix}$

$\Phi(t,\tau) = \exp\left(\int_{\tau}^{t} A(s)\,ds \right)$

$\int_{\tau}^{t} A(s)\,ds = \int_{\tau}^{t} \begin{bmatrix} -1 & 2s \\ 0 & -1 \end{bmatrix} ds = \begin{bmatrix} -(t-\tau) & t^2 - \tau^2 \\ 0 & -(t-\tau) \end{bmatrix}$ — (1)

let $t - \tau = e$ & $f = t^2 - \tau^2$

we see, $\int_{\tau}^{t} A(s) = e \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} + 2s \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = -I + 2sN$

(1) become $\begin{bmatrix} -e & f \\ 0 & -e \end{bmatrix} = -e \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + f \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$= -eI + fN$

$= -(t-\tau)I + (t^2 - \tau^2)N$

$\therefore \Phi(t,\tau) = e^{-(t-\tau)I} \cdot e^{(t^2-\tau^2)N}$ since $-(t-\tau)I$ & $(t^2-\tau^2)N$ commute.

$e^{-(t-\tau)I} = e^{-(t-\tau)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$e^{(t^2-\tau^2)N} = I + (t^2-\tau^2)N + \frac{1}{2!}(t^2-\tau^2)N^2 + \cdots = \begin{bmatrix} 1 & t^2-\tau^2 \\ 0 & 1 \end{bmatrix}$

$\therefore \Phi(t,\tau) = \begin{bmatrix} e^{-(t-\tau)} & (t^2-\tau^2)\,e^{-(t-\tau)} \\ 0 & e^{-(t-\tau)} \end{bmatrix}$ Answer

3(b) $x(t)$ given $x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $u(t) = e^{-t}$, $u(\tau) = e^{-\tau}$

$\tau = 0$ in $\phi(t, \tau)$

$$x(t) = \underbrace{\phi(t,0)\,x(0)}_{ZIR} + \underbrace{\int_0^t \phi(t,\tau)\,Bu(\tau)\,d\tau}_{ZSR}$$

$$x(t,0) = \begin{bmatrix} e^{-t} & t^2 e^{-t} \\ 0 & e^{-t} \end{bmatrix}$$

$$\therefore ZIR = \begin{bmatrix} e^{-t} & t^2 e^{-t} \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^{-t} \\ 0 \end{bmatrix}$$

$$\phi(t,\tau)B = \begin{bmatrix} e^{-(t-\tau)} & (t^2-\tau^2)\,e^{-(t^2-\tau^2)} \\ 0 & e^{-(t-\tau)} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} (t^2-\tau^2)\,e^{-(t-\tau)} \\ e^{-(t-\tau)} \end{bmatrix}$$

$$\phi(t,\tau)\,u(\tau) = \begin{bmatrix} (t^2-\tau^2)\,e^{-t} \\ e^{-t} \end{bmatrix}$$

$$\therefore \int_0^t \begin{bmatrix} \quad \end{bmatrix} d\tau$$

each row $e^{-t}\int_0^t (t^2-\tau^2)\,d\tau = e^{-t}\left[t^2\tau - \frac{\tau^3}{3}\right]_0^t$

$$= e^{-t}\left[t^3 - \frac{t^3}{3}\right] = \frac{2}{3}t^3 e^{-t}$$

& $\int_0^t e^{-t}\,d\tau = te^{-t}$

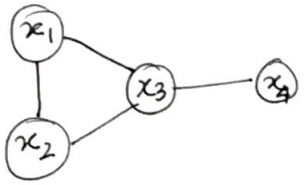
$$\therefore ZSR = \int_0^t \phi(t,\tau)\,Bu(\tau)\,d\tau = \begin{bmatrix} 2/3\,t^3 e^{-t} \\ te^{-t} \end{bmatrix}$$

$$\therefore x(t) = ZIR + ZSR$$

$$= \begin{bmatrix} e^{-t} \\ 0 \end{bmatrix} + \begin{bmatrix} 2/3\,t^3 e^{-t} \\ te^{-t} \end{bmatrix} = \begin{bmatrix} (1 + \frac{2}{3}t^3)\,e^{-t} \\ te^{-t} \end{bmatrix}$$

Any

(P4)



$$x[k+1] = Ax[k] + Bu[k] = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} x[k] + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} u[k] \quad \text{——— (1)}$$

$k = 0 \quad x[0] = 0$

(a) $k = 0, 1, 2, 3 \;\rightarrow x[4] \in \mathbb{R}^4$

$k=0, \quad x[1] = Ax[0] + Bu[0] = Bu[0]$

$k=1 \quad x[2] = Ax[1] + Bu[1] = A.Bu[0] + Bu[1]$

$k=2 \quad x[3] = Ax[2] + Bu[2] = A.(A.Bu[0] + Bu[1]) + Bu[2]$

$= A^2 Bu[0] + AB u[1] + Bu[2]$

$= A^2 Bu[0] + AB u[1] + Bu[2]$

$k=3 \quad x[4] = Ax[3] + Bu[3]$

$= A\left(A^2 Bu[0] + AB u[1] + Bu[2]\right) + Bu[3]$

$= A^3 Bu[0] + A^2 Bu[1] + AB u[2] + Bu[3]$ Ans

$\therefore x[4] = A^3 Bu[0] + A^2 Bu[1] + AB u[2] + Bu[3]$

matrix form

$$x[4] = \begin{bmatrix} A^3B & A^2B & AB & B \end{bmatrix} \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[3] \end{bmatrix}$$ Ans

$\Downarrow F$

$$x[4] = F \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[3] \end{bmatrix}$$

where $F = \begin{bmatrix} A^3B & A^2B & AB & B \end{bmatrix}$

Given $B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\therefore AB = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/4 \\ 0 \end{bmatrix}$

$$A^2B = A\cdot(AB) = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 1/3 \\ 1/3 \\ 1/4 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3\cdot 1/3 + \frac{1}{3}\cdot\frac{1}{3} + \frac{1}{3}\cdot\frac{1}{12} + 0 \\ 1/9 + 1/9 + 1/12 + 0 \\ 1/12 + 1/12 + 1/16 + 0 \\ 0 + 0 + 1/8 + 0 \end{bmatrix} = \begin{bmatrix} 1/9 + \frac{1}{9} + \frac{1}{12} \\ 1/9 + \frac{1}{9} + \frac{1}{12} \\ 1/12 + 1/12 + \frac{1}{16} \\ 1/8 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{12+12+9}{108} \\ \frac{12+12+9}{108} \\ \frac{4+4+3}{48} \\ 1/8 \end{bmatrix} = \begin{bmatrix} 11/36 \\ 11/36 \\ 11/48 \\ 1/8 \end{bmatrix}$$

$$A^3B = A(A^2B) = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 11/36 \\ 11/36 \\ 11/48 \\ 1/8 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3}\cdot\frac{11}{36} + \frac{1}{3}\cdot\frac{11}{36} + \frac{1}{3}\cdot\frac{11}{48} \\ \frac{1}{3}\cdot\frac{11}{36} + \frac{1}{3}\cdot\frac{11}{36} + \frac{1}{3}\cdot\frac{11}{48} \\ \frac{1}{4}\cdot\frac{11}{36} + \frac{1}{4}\cdot\frac{11}{36} + \frac{1}{4}\cdot\frac{11}{48} + \frac{1}{4}\frac{11}{48} \\ 0 + 0 + \frac{1}{2}\cdot\frac{11}{48} + \frac{1}{2}\cdot\frac{1}{8} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{11}{108} + \frac{11}{108} + \frac{11}{144} \\ \frac{11}{108} + \frac{11}{108} + \frac{11}{144} \\ 11/144 + 11/144 + \frac{11}{192} + \frac{1}{32} \\ \frac{11}{96} + \frac{1}{16} \end{bmatrix} = \begin{bmatrix} \frac{144+144+108}{1552} \\ \frac{144+144+108}{1552} \\ \frac{884736\times 3 + 3981312}{12740198\,4} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3}\left[\frac{11}{36} + \frac{1}{36} + \frac{11}{48}\right] \\ \frac{1}{3}\left[\frac{11}{36} + \frac{11}{36} + \frac{11}{48}\right) \\ \frac{1}{4}\left(\frac{11}{36} + \frac{11}{36} + \frac{11}{48} + \frac{1}{8}\right) \\ \frac{1}{2}(11/48 + 1/8) \end{bmatrix} = \begin{bmatrix} \frac{1}{3}\cdot\left(\frac{44+44+33}{144}\right) \\ 1/3\left(\frac{44+44+33}{144}\right) \\ \frac{1}{4}\left(\frac{44+44+33+18}{144}\right) \\ \frac{1}{2}\left(\frac{11+6}{48}\right) \end{bmatrix}$$

$$A^3B = \begin{bmatrix} \frac{1}{3}\left(\frac{121}{144}\right) \\ \frac{1}{3}\left(\frac{121}{144}\right) \\ \frac{1}{4}\left(\frac{139}{144}\right) \\ \frac{1}{2}\left(\frac{17}{48}\right) \end{bmatrix} = \begin{bmatrix} \frac{121}{432} \\ \frac{121}{432} \\ \frac{139}{576} \\ \frac{17}{96} \end{bmatrix}$$

$$\therefore F = \begin{bmatrix} A^3B & A^2B & AB & B \\ \frac{121}{432} & 11/36 & 1/3 & 1 \\ \frac{121}{432} & 11/36 & 1/3 & 0 \\ \frac{139}{576} & 11/48 & 1/4 & 0 \\ \frac{17}{96} & 1/8 & 0 & 0 \end{bmatrix}$$

[Apparently could use Matlab :]

Answer

$\therefore x[4] = F \begin{bmatrix} u[0] & u[1] & u[2] & u[3] \end{bmatrix}^T$

P4

(b) Given new target $x[4] = [-3 \quad -1 \quad 1 \quad 3]^T$ $\therefore F.u = x[4]$ here,

$$\begin{bmatrix} \frac{121}{432} & 11/36 & 1/3 & 1 \\ \frac{121}{432} & 11/36 & 1/3 & 0 \\ \frac{139}{576} & 11/48 & 1/4 & 0 \\ 17/96 & 1/8 & 0 & 0 \end{bmatrix} \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[3] \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

F ↑ u ↑ x[4] new target.

Using matlab (Code attached) $\det[F] = -\frac{1}{768} < 0$ so non singular

$\therefore$ Rank[F] = 4 & full rank so can be inverted

Since non singular & invertible I think it can find target $x[4]$

$$\therefore \frac{121}{432}u[0] + \frac{11}{36}u[1] + \frac{1}{3}u[2] + u[3] = -3 \quad \text{—(1)}$$

$$\frac{121}{432}u[0] + \frac{11}{36}u[1] + \frac{1}{3}u[2] + 0.u[3] = -1 \quad \text{—(2)}$$

$$139/576\,u[0] + 11/48\,u[1] + 1/4\,u[3] + 0.u[3] = 1 \quad \text{—(3)}$$

$$17/96\,u[0] + 1/8\,u[1] + 0.u[2] + 0.u[3] = 3 \quad \text{—(4)}$$

② − ① ⇒ $u[3] = -2$ — (5)

~~③ − ⑥~~ ② − ③

⑧ − ⑩ $+ \frac{121}{432} u[0] + \frac{11}{36} u[1] + \frac{1}{3} u[2] - \frac{139}{576} u[0] - \frac{11}{48} u[1] - \frac{1}{4} u[2]$

$= +121\,u[0] + 132\,u[1] + 144\,u[2] - 139\,u[0] - 132\,u[1] - 144\,u[2]$

$= -432 - 576$

or $-18\,u[0] = -1008$

$u[0] = \frac{1008}{18} = 56$

$\therefore u[0] = 56$ — (6)

from (4), add $u[0]$: $\frac{17}{96} \cdot 56 + \frac{1}{8} u[1] = 3$

$952 + \frac{12}{8} u[1] = 288$

$12\,u[1] = -664$

$u[1] = -\frac{166}{3}$

$u[1] = -\frac{166}{3}$

from (2) ~~$\frac{121}{432} \cdot 56 + \frac{11}{36} u[1] + \frac{1}{3}$~~

$\frac{121}{432} \cdot 56 + \frac{11}{36} \cdot \left(-\frac{166}{3}\right) + \frac{1}{3} u[2] = -1$

$121 \cdot 56 + 132 \left(-\frac{166}{3}\right) + 144\,u[2] = -432$

$144\,u[2] = -432 + 528$

$= 96$

$u[2] = \frac{96}{144} = 2/3$

$\therefore$ → Ans is the answer to find input sequence.

$$u = \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[3] \end{bmatrix} = \begin{bmatrix} 56 \\ -166/3 \\ 2/3 \\ -2 \end{bmatrix}$$

Answer

P4 (c) $B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ repeat ?? ∴

Same as part (a) $x[4] = A^3Bu[0] + A^2Bu[1] + ABu[2] + Bu[3]$.

$\therefore \quad F_2 = \begin{bmatrix} A^3B & A^2B & AB & B \end{bmatrix}$

$$AB = \begin{bmatrix} 1/3 & -1/3 & 1/3 & 0 \\ 1/3 & 1/3 & -1/3 & 0 \\ -1/4 & 1/4 & 1/4 & -1/4 \\ 0 & 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1/4 \\ -1/2 \end{bmatrix}$$

$$A^2B = A(AB) = \begin{bmatrix} 1/3 & -1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ -1/4 & 1/4 & 1/4 & -1/4 \\ 0 & 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1/4 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 1/12 \\ -1/12 \\ 3/16 \\ 3/8 \end{bmatrix}$$

$$A^3B = A(A^2B) = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{bmatrix} -1/12 \\ -1/12 \\ 3/16 \\ 3/8 \end{bmatrix} = \begin{bmatrix} \frac{1}{36} + \frac{1}{36} + \frac{3}{48} \\ -1/36 - 1/36 + 3/48 \\ -1/48 - 1/48 + \frac{3}{64} + \frac{3}{32} \\ 0 + 0 + \frac{1}{8} + \frac{1}{8} \end{bmatrix}$$

$$= \begin{bmatrix} 17/144 \\ 17/144 \\ 35/192 \\ 9/32 \end{bmatrix}$$

$$F_2 = \begin{bmatrix} \frac{17}{144} & 1/12 & 0 & 0 \\ \frac{17}{144} & -1/12 & 0 & 0 \\ \frac{35}{192} & 3/16 & 1/4 & 0 \\ 9/32 & 3/8 & 1/2 & 1 \end{bmatrix} \quad \therefore \quad x[4] = F_2 \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[3] \end{bmatrix}$$

now $F_2 \cdot u = x[4]$ targeting evaluation

$$\begin{bmatrix} \frac{17}{144} & 1/12 & 0 & 0 \\ \frac{17}{144} & -1/12 & 0 & 0 \\ \frac{35}{92} & 3/16 & 1/4 & 0 \\ 9/32 & 3/8 & 1/2 & 1 \end{bmatrix} \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ u[4] \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

here det = 0 from matlab.

and rank $(F_2) = 3$

```
clear;
clc;

A = [sym(1)/3, sym(1)/3, sym(1)/3, sym(0);
    sym(1)/3, sym(1)/3, sym(1)/3, sym(0);
    sym(1)/4, sym(1)/4, sym(1)/4, sym(1)/4;
    sym(0),   sym(0),   sym(1)/2, sym(1)/2];

% Part (a): input at sensor 1
B = [sym(1); sym(0); sym(0); sym(0)];
F = [A^3*B, A^2*B, A*B, B];

disp('F =')
```

```
F =
```

```
disp(F)
```

$$\begin{pmatrix} \frac{121}{432} & \frac{11}{36} & \frac{1}{3} & 1 \\ \frac{121}{432} & \frac{11}{36} & \frac{1}{3} & 0 \\ \frac{139}{576} & \frac{11}{48} & \frac{1}{4} & 0 \\ \frac{17}{96} & \frac{1}{8} & 0 & 0 \end{pmatrix}$$

```
detF = simplify(det(F))
```

```
detF =
```

$$-\frac{1}{768}$$

```
rankF = rank(F)
```

```
rankF =
4
```

```
% Part (c): input at sensor 4
D = [sym(0); sym(0); sym(0); sym(1)];
F4 = [A^3*D, A^2*D, A*D, D];

disp('F4 =')
```

```
F4 =
```

```
disp(F4)
```

$$\begin{pmatrix} \frac{17}{144} & \frac{1}{12} & 0 & 0 \\ \frac{17}{144} & \frac{1}{12} & 0 & 0 \\ \frac{35}{192} & \frac{3}{16} & \frac{1}{4} & 0 \\ \frac{9}{32} & \frac{3}{8} & \frac{1}{2} & 1 \end{pmatrix}$$

```
detF4 = simplify(det(F4))
```

detF4 = 0

```
rankF4 = rank(F4)
```

```
rankF4 =
3
```

Row 1 & Row 2 of F_2 are same

$\therefore \quad x_1[4] = \frac{17}{144} u[0] + \frac{1}{12} u[1]$

& $x_2[4] = \frac{17}{144} u[0] + \frac{1}{12} u[1]$

so $x_1[4] = x_2[4]$ and target needed is $x_1 = -3$ & $x_2 = -1$

but here $x_1 = x_2 \neq$ target values as $-3 \neq -1$

$\therefore$ So can't be satisfied also rank = 3 $< \mathbb{R}^4$

& determinant is zero so singular non invertible so matrix can't reach target if $B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Ans

— x —

P5 CTLTI

$$\dot{x} = Ax + Bu = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

A B

Given, $x(0) = 0 \quad i.e \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

a) $\dot{x}_1 = -x_1 + x_2 \quad \text{---} (1) \qquad u(t) = 1$

$\dot{x}_2 = 0 \cdot x_1 - 2x_2 + 1 \cdot \quad \text{---} (2)$

from (2) $\therefore \dot{x}_2 + 2x_1 = 1.$ IF: $\mu(t) = e^{\int 2\,dt} = e^{2t}$

$e^{2t} \dot{x}_2 + 2e^{2t} x_1 = e^{2t}$

$\frac{d}{dt} \left(e^{2t} \cdot x_2(t) \right) = e^{2t}$

$\int_0^t d\left(e^{2t} x_2(t) \right) = \int_0^t e^{2t}\, dt$

$= e^{2t} x_2(t) - x_2(0) = \frac{1}{2} e^{2t} \Big]_0^t = \frac{1}{2} e^{2t} - \frac{1}{2}$

$\therefore \quad e^{2t} x_2(t) = \frac{1}{2} e^{2t} - \frac{1}{2}$

$\therefore \quad x_2(t) = \frac{1}{2} - \frac{1}{2} e^{-2t} \quad \text{---} (3)$

(3) into (1)

$$\dot{x}_1 + x_1 = \frac{1}{2} - \frac{1}{2} e^{-2t}$$

IF $\mu(t) = e^{\int dt} = e^t$

$$e^t \cdot \dot{x}_1 + e^t x_1 = \frac{1}{2} e^t - \frac{1}{2} \frac{e^t}{e^{2t}}$$

$$\frac{d}{dt}\left(e^t x_1(t)\right) = \frac{1}{2} e^t - \frac{1}{2} e^{-t}$$

$$\int_0^t d\left(e^t x_1(t)\right) = \int_0^t \left(\frac{1}{2} e^t - \frac{1}{2} e^{-t}\right) dt$$

$$e^t x_1(t) - x_1(0) = \left[\frac{1}{2} e^t + \frac{1}{2} e^{-t}\right]_0^t = \frac{1}{2} e^t - \frac{1}{2} + \frac{1}{2} e^{-t} - \frac{1}{2}$$

$$= \frac{1}{2}\left(e^t + e^{-t}\right) - 1$$

$$\therefore \quad x_1(t) = e^{-t}\left(\frac{1}{2}\left(e^t + e^{-t}\right) - 1\right)$$

$$= \frac{1}{2} + \frac{1}{2} e^{-2t} - e^t$$

$$\therefore \quad x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2} e^{-2t} \\ \frac{1}{2} - \frac{1}{2} e^{-2t} \end{bmatrix}$$ Ans

The plot is provided in mattab attached.

P5 (b)

$T = 0.5$ sec.

$$x[k+1] = A_d x[k] + B_d u[k]$$

$$x[k+1] = \underbrace{e^{AT}}_{A_d} x[k] + \underbrace{\left(\int_0^T e^{A\tau} d\tau\right) B}_{B_d} u[k]$$

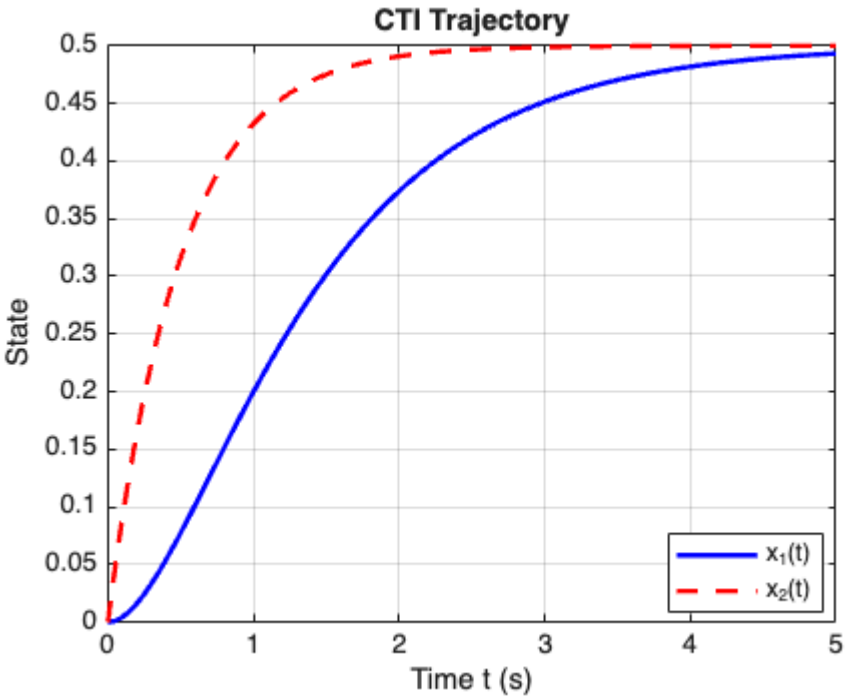
$$A_d = e^{AT} = e^{A(0.5)} = \begin{bmatrix} & \\ & \end{bmatrix}$$

```
t = linspace(0, 5, 501);

x1 = 0.5 - exp(-t) + 0.5*exp(-2*t);
x2 = 0.5 - 0.5*exp(-2*t);

plot(t, x1, 'b-', 'LineWidth', 1.8);
hold on;
plot(t, x2, 'r--', 'LineWidth', 1.8);
grid on;

xlabel('Time t (s)');
ylabel('State');
legend('x_1(t)', 'x_2(t)', 'Location', 'best');
title('CTLTI Trajectory');
xlim([0 5]);
```



$A: \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$

$-x_1 + x_2 = 0$

$-2x_2 = 0$

$= \begin{bmatrix} -1-\lambda & 1 \\ 0 & -2-\lambda \end{bmatrix} = 0$

$+(1-\lambda)(2-\lambda) - 0 = 0$

$\therefore \lambda = -1 \;,\; \lambda = -2$

$x_1 + x_2 = 0$

$\lambda = -1$

$x_1 = 1$

$x_2 = 0$

$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda = -2$

$x_1 + x_2 = 0$

$x_1 = -x_2$

$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$T: \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \quad T^{-1} = \frac{1}{-1-0} \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} =$

$= \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = T$

$e^{At} = T e^{\Lambda t} T^{-1}$

$e^{\Lambda t} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix}$

$\therefore e^{At} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^{-t} & e^{-t} \\ 0 & -e^{-2t} \end{bmatrix}$

$= \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix} \leftarrow$

$= \begin{bmatrix} e^{-0.5} & e^{-0.5} - e^{-1.0} \\ 0 & e^{-1.0} \end{bmatrix}$

Next,

$\int_0^{0.5} \left(e^{A\tau} d\tau \right) B \qquad e^{A\tau} B = \begin{bmatrix} e^{-\tau} & e^{-\tau} - e^{-2\tau} \\ 0 & e^{-2\tau} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} e^{-\tau} - e^{-2\tau} \\ e^{-2\tau} \end{bmatrix}$

$\therefore \int_0^T e^{A\tau} B \, d\tau = \begin{bmatrix} \int_0^T (e^{-\tau} - e^{-2\tau}) d\tau \\ \int_0^T e^{-2\tau} d\tau \end{bmatrix} = \begin{bmatrix} \left[-e^{-\tau} + \frac{1}{2} e^{-2\tau} \right]_0^T \\ \left[-\frac{1}{2} e^{-2\tau} \right]_0^T \end{bmatrix}$

$$\int_0^T e^{A\tau} B\, d\tau = \begin{bmatrix} -e^{-T} + \frac{1}{2}e^{-2T} + 1 - \frac{1}{2} \\ -\frac{1}{2}e^{-2T} + \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} - e^{-T} + \frac{1}{2}e^{-2T} \\ \frac{1}{2} - \frac{1}{2}e^{-2T} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} - e^{-0.5} + \frac{1}{2}e^{-1.0} \\ \frac{1}{2} - \frac{1}{2}e^{-1.0} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-0.5} + \frac{1}{2}e^{-1.0} \\ \frac{1}{2} - \frac{1}{2}e^{-1.0} \end{bmatrix}$$

$$\therefore x[k+1] = \underbrace{e^{AT}}_{A_d} x[k] + \underbrace{\int_0^T e^{A\tau}\, d\tau\, B}_{B_d} u[k]$$

$$= \begin{bmatrix} e^{-0.5} & e^{-0.5} - e^{-1.0} \\ 0 & e^{-1.0} \end{bmatrix} x[k] + \begin{bmatrix} \frac{1}{2} - e^{-0.5} + \frac{1}{2}e^{-1.0} \\ \frac{1}{2} - \frac{1}{2}e^{-1.0} \end{bmatrix} u[k]$$

Ans

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(c) The plot & code attached below.

From part (a) $x(t) = \begin{bmatrix} \frac{1}{2} & e^{-t} + \frac{1}{2}e^{-2t} \\ & \frac{1}{2} - \frac{1}{2}e^{-2t} \end{bmatrix}$

$T = 0.5$, $t = kT = 0.5k$ give

$$x(kT) = \begin{bmatrix} \frac{1}{2} - e^{-0.5k} + \frac{1}{2}e^{-k} \\ \frac{1}{2} - \frac{1}{2}e^{-k} \end{bmatrix} \quad k = 0, 1, \ldots, 10$$

The matlab output X shows the discrete states $x[0], \ldots, x[10]$ & Xcont show CT states evaluated at corresponding times kT. The max error ≈ 0 so

$$x[k] = x(kT) \quad \text{for } k = 0, 1, \ldots, 10$$

Answer

```
clear;
clc;
close all;

% CTI: x_dot = A*x + B*u
A = [-1  1;
    0 -2];

B = [0; 1];

% Sampling info
T = 0.5;
N = 10;
k = 0:N;
tk = k*T;

% zero-order-hold
Ad = expm(A*T);
Bd = A \ ((Ad - eye(2))*B);

disp('Ad =');
```

```
Ad =
```

```
disp(Ad);
```

```
    0.6065    0.2387
         0    0.3679
```

```

disp('Bd =');
```

```
Bd =
```

```
disp(Bd);
```

```
    0.0774
    0.3161
```

```

% input u[k] = 1
u = ones(1, N);

% x[k+1] = Ad*x[k] + Bd*u[k], with x[0] = [0; 0]

X = zeros(2, N+1);
for n = 1:N
    X(:,n+1) = Ad*X(:,n) + Bd*u(n);
end

% Display
disp('Cols of X are x[0], x[1], ..., x[10]:');
```

Cols of X are x[0], x[1], ..., x[10]:

```
disp(X);
```

```
         0    0.0774    0.1998    0.3018    0.3738    0.4213    0.4515    0.4703    0.4819    0.4890    0.
         0    0.3161    0.4323    0.4751    0.4908    0.4966    0.4988    0.4995    0.4998    0.4999    0.
```

```
% CT response from part (a)
Xcont = [0.5 - exp(-tk) + 0.5*exp(-2*tk);
    0.5 - 0.5*exp(-2*tk)];

% Verify x[k] = x(kT)
difference = X - Xcont;
maxError = max(abs(difference(:)));

disp('Max difference between x[k] and x(kT):');
```

Max difference between x[k] and x(kT):

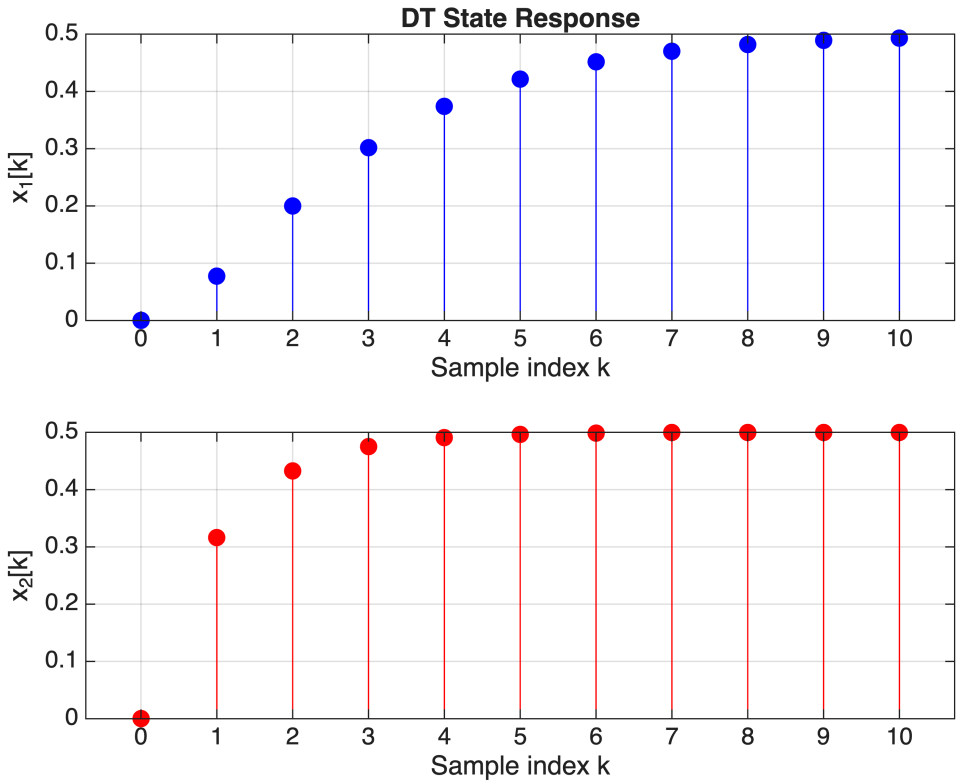
```
disp(maxError);
```

```
   5.5511e-17
```

```
%% Plot 1: DT states only
figure;

subplot(2,1,1);
stem(k, X(1,:), 'b', 'filled');
grid on;
xlabel('Sample index k');
ylabel('x_1[k]');
title('DT State Response');

subplot(2,1,2);
stem(k, X(2,:), 'r', 'filled');
grid on;
xlabel('Sample index k');
ylabel('x_2[k]');
```

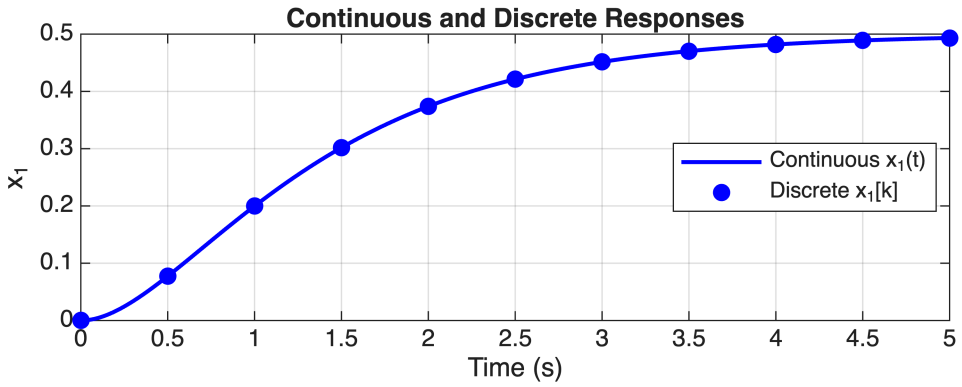


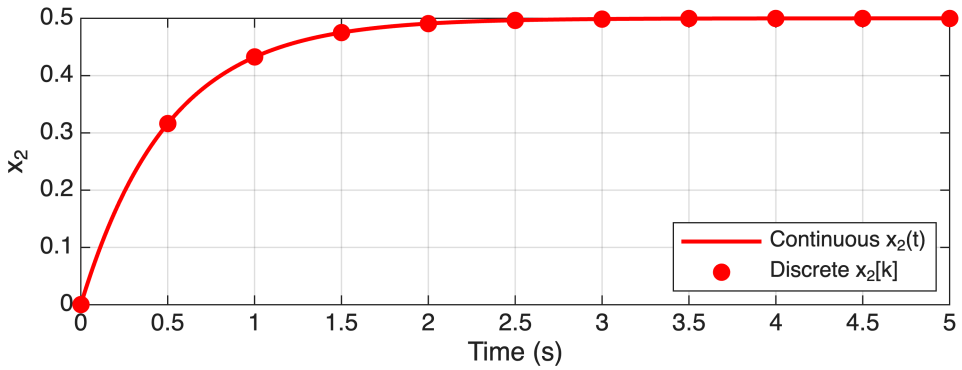

```
%% Plot 2: CT response with discrete samples
tFine = linspace(0, 5, 501);
x1Fine = 0.5 - exp(-tFine) + 0.5*exp(-2*tFine);
x2Fine = 0.5 - 0.5*exp(-2*tFine);

figure;

subplot(2,1,1);
plot(tFine, x1Fine, 'b-', 'LineWidth', 1.5);
hold on;
plot(tk, X(1,:), 'bo', 'MarkerFaceColor', 'b');
grid on;
xlabel('Time (s)');
ylabel('x_1');
legend('Continuous x_1(t)', 'Discrete x_1[k]', 'Location', 'best');
title('Continuous and Discrete Responses');

subplot(2,1,2);
plot(tFine, x2Fine, 'r-', 'LineWidth', 1.5);
hold on;
plot(tk, X(2,:), 'ro', 'MarkerFaceColor', 'r');
grid on;
xlabel('Time (s)');
ylabel('x_2');
legend('Continuous x_2(t)', 'Discrete x_2[k]', 'Location', 'best');
```


Continuous and Discrete Responses
x_1
Time (s)
Continuous $x_1(t)$
Discrete $x_1[k]$


x_2
Time (s)
Continuous $x_2(t)$
Discrete $x_2[k]$